

Merve Ileri-Tayar

Postdoctoral Research Associate

Department of Psychology & Neuroscience, Duke University

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Employment

2026- Postdoctoral Research Associate
Department of Psychology & Neuroscience
Duke University, NC, USA

Education

2021-2026 Ph.D., Behavior, Brain, & Cognition
Washington University in St. Louis, MO, USA
GPA: 4.00/4.00
Advisor: Dr. Julie M. Bugg
Dissertation Title: *Learning and Transferring Value in Conflict Tasks: The Architecture of Reward-Based Attentional Priorities*

2021-2023 M.A., Behavior, Brain, & Cognition
Washington University in St. Louis, MO, USA
GPA: 4.00/4.00
Advisor: Dr. Julie M. Bugg
Master's Thesis Title: *Between-Task Transfer of Learned Control Settings: How Far Can It Go?*

2018-2021 M.Sc., Cognitive Psychology
Middle East Technical University, Ankara, Turkey
GPA: 4.00/4.00
Advisors: Dr. Mine Misirlisoy, Dr. Nart Bedin Atalay
Master's Thesis Title: *The Role of Expectation and Experience in the Control of Attention*

2014-2018 B.A., Psychology
TOBB University of Economics and Technology, Ankara, Turkey
GPA: 3.79/4.00

Honors

2021-2026 McDonnell International Scholars Academy Fellowship
2026 Commencement Ceremony Selected Graduate Student Speaker, Washington University in St. Louis

2026	McDonnell International Scholars Academy PhD-in-3D Competition (Runner-up)
2025	Dean's Award for Graduate Research Excellence (Winner), Washington University in St. Louis
2025	Departmental Dissertation Grant, Washington University in St. Louis
2024	Psychonomic Society Graduate Travel Award
2022	Psychonomic Society J. Frank Yates Student Conference Award
2021	Fulbright Ph.D. Grant (Awarded, Declined)
2020-2021	Middle East Technical University M.Sc. Thesis Project Grant
2018-2021	2210-A National Scholarship Program for M.Sc. Students, The Scientific and Technological Research Council of Turkey (TUBİTAK)
2014-2018	Full-Merit Scholarship, Department of Psychology, TOBB University of Economics and Technology, Turkey

Publications

- Ileri-Tayar, M.**, Bugg, J. M., & Kool, W. (2026). Switching between cognitive control states? No, thank you. *Journal of Experimental Psychology: General*, 155(7), 1726-1747. <https://doi.org/10.1037/xge0001940>
- Ileri-Tayar, M.**, Bugg, J. M., Dudey, T., & Braver, T. S. (2025). Proactive control declines while reactive control is preserved across the adult lifespan. *Journal of Experimental Psychology: General*, 154(11), 3029-3047. <https://psycnet.apa.org/doi/10.1037/xge0001824>
- Nuño, C., **Ileri-Tayar, M.**, & Bugg, J. M. (2025). Attention on demand: Do people strategically heighten control when distraction is expected but rare? *Psychological Research*, 89, 106. <https://doi.org/10.1007/s00426-025-02131-2>
- Ileri-Tayar, M.**, Suh, J., Stern, A., Whitsitt, L., & Bugg, J. M. (2025). Does an external distractor interfere with the triggering of item-specific control? *Journal of Experimental Psychology: Human Perception and Performance*, 51(6), 808–825. <https://doi.org/10.1037/xhp0001323>
- Ileri-Tayar, M.**, Colvett, J. S., Dey, A., & Bugg, J. M. (2025). Does “item-specific” cognitive control operate at the item level? *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 51(8), 1238–1258. <https://doi.org/10.1037/xlm0001432>
- Ileri-Tayar, M.**, Colvett, J. S., & Bugg, J. M. (2024). Between-task transfer of item-specific control is replicable and extends to novel conditions. *Journal of Experimental Psychology: Human Perception and Performance*, 50(6), 535-553. <https://doi.org/10.1037/xhp0001200>
- Ileri-Tayar, M.**, Moss, C., & Bugg, J. M. (2022). Transfer of learned cognitive control settings within and between tasks. *Neurobiology of Learning and Memory*, 196, 107689. <https://doi.org/10.1016/j.nlm.2022.107689>
- Suh, J., **Ileri-Tayar, M.**, & Bugg, J. M. (2022). When global and local information about attentional demands collide: evidence for global dominance. *Attention, Perception, & Psychophysics*, 84(6), 1858-1873. <https://doi.org/10.3758/s13414-022-02521-3>

Manuscripts Submitted for Publication

- Ileri-Tayar, M.**, & Bugg, J. M. (under review). Learning and transfer of item-reward associations in conflict tasks. *Journal of Experimental Psychology: Human Perception and Performance*.

Manuscripts in Preparation

Ileri-Tayar, M., Dey, A., Kool, W., & Bugg, J. M. (in preparation). Reward-induced shifts in cognitive control.

Ileri-Tayar, M., Dreisbach, G., & Bugg, J. M. (in preparation). Age-related changes in task-switching: Distinct underlying mechanisms.

Ileri-Tayar, M., & Bugg, J. M. (in preparation). The puzzle of context-level reward learning.

Invited Talks and Talks at Conferences

Ileri-Tayar, M., & Bugg, J. M. (March, 2026). Within- and between-task transfer of item-reward associations in conflict tasks. Talk presented at 68th TeaP (Conference of Experimental Psychologists). Tübingen, Germany.

Ileri-Tayar, M. & Bugg, J. M. (March, 2026). Learning and Transfer of Adaptive Cognitive Control. Invited talk at Rothermund Lab, Friedrich Schiller University Jena, Germany.

Ileri-Tayar, M., Dey, A., Bugg, J. M., & Kool, W. (November, 2025). Target enhancement or distractor suppression? Modeling the source of reward-induced control enhancements. Talk presented at the Future of Cognitive Control Meeting. Denver, Colorado.

Ileri-Tayar, M., Bugg, J. M., & Kool, W. (September, 2025). Switching between cognitive control states? No, thank you. Talk presented at the 24th Biennial Conference of the European Society for Cognitive Psychology. Sheffield, United Kingdom.

Bugg, J. M., & **Ileri-Tayar, M.** (July, 2025). The automaticity and flexibility of learning-guided item-specific control. Invited talk presented virtually at the Symposium on Conflict Adaptation across Single- and Multi-Tasking Paradigms, RWTH Aachen University, Aachen, Germany.

Ileri-Tayar, M., Kool, W., & Bugg, J. M. (November, 2024). Reward induces shifts in learning-guided reactive control. Talk presented at the 65th Annual Meeting of the Psychonomic Society. New York City, New York.

Ileri-Tayar, M. & Bugg, J. M. (May, 2024). Between-task transfer of learned control settings: How far can it go? Invited talk at Dreisbach Lab, University of Regensburg, Germany.

Ileri-Tayar, M. & Bugg, J. M. (December, 2023). Between-task transfer of learned control settings. Invited talk at Braem Lab, Ghent University, Belgium.

Atalay, N. B., Bozkurt, O., & **Ileri-Tayar, M.** (October, 2022). Investigation of list-wide proportion congruence effect with response tracking methods. Talk presented at the 21st National Psychology Congress, Istanbul, Turkey.

Ileri, M., & Atalay, N. B. (November, 2018). Is asymmetrical list-shifting effect observed with item-specific shifting? Talk presented at the 20th National Psychology Congress, Ankara, Turkey.

Conference Poster Presentations

Ileri-Tayar, M., & Bugg, J. M. (November, 2025). When more information hurts: explicit reward information prevents transfer. Poster presented at the 66th Annual Meeting of the Psychonomic Society. Denver, Colorado.

Ileri-Tayar, M., Bugg, J. M., & Kool, W. (November, 2024). Switching between cognitive control states? No, thank you. Poster presented at the 65th Annual Meeting of the Psychonomic Society. New York City, New York.

- Ileri-Tayar, M.**, Moss, C., & Bugg, J. M. (March, 2024). Does reward induce shifts in learning-guided reactive control? Poster presented at the 66th TeaP (Conference of Experimental Psychologists). Regensburg, Germany.
- Ileri-Tayar, M.**, Bugg, J. M., & Braver, T. S. (November, 2023). A within-subjects comparison dissociating proactive and reactive control in younger, middle-aged, and older adults. Poster presented at the 64th Annual Meeting of the Psychonomic Society. San Francisco, California.
- Ileri-Tayar, M.**, Moss, C., & Bugg, J. M. (November, 2022). Transfer of learned cognitive control settings within and between tasks. Poster presented at the 63rd Annual Meeting of the Psychonomic Society. Boston, Massachusetts.
- Ic, Y., **Ileri-Tayar, M.**, & Atalay, N. B. (October, 2022). The influence of expectation on list-shifting: Initial findings. Poster presented at the 21st National Psychology Congress, Istanbul, Turkey.
- Ileri, M.**, Misirlisoy, M., & Atalay, N. B. (May, 2020). Does response type change the Stroop effect in a mouse-tracking experiment? Poster presented virtually at the International Symposium on Brain and Cognitive Science.
- Ileri, M.**, Misirlisoy, M., & Atalay, N. B. (September, 2019). Investigation of the Stroop effect with response tracking: Influence of response type. Poster presented at the 5th Experimental and Cognitive Psychology Symposium, Istanbul, Turkey.

Teaching Experiences

2024	Instructor, Washington University in St. Louis, USA Introduction to R (Summer course)
2023-2024	Assistant in Instruction, Washington University in St. Louis, USA Graduate courses: Quantitative Methods 1, Quantitative Methods 2
2019-2021	Teaching Assistant, Middle East Technical University, Turkey Undergraduate courses: Open Science and Replicability, Topics in Experimental Psychology: Brain and Behavior, Experimental Design & Analysis, Memory Graduate courses: Experimental Design & Analysis

Pre-Ph.D. Research Experiences

2019-2021	Lab Coordinator Attention and Memory Lab, Middle East Technical University
2019-2021	Project Research Assistant The Scientific and Technological Research Council of Turkey (TUBİTAK), 1001 Support Program for Scientific and Technological Research Projects
2018-2019	Lab Coordinator

Cognition Lab, TOBB University of Economics and Technology

2017-2021 Research Assistant
Attention and Memory Lab, Middle East Technical University

2017-2019 Research Assistant
Cognition Lab, TOBB University of Economics and Technology

Programming & Software Knowledge

R	JavaScript	High-performance
Python	E-Prime	computing (HPC) clusters

Professional Memberships

Graduate Student Member, Psychonomic Society
Graduate Student Member, American Psychological Association
Associate Member, European Society for Cognitive Psychology
Women+ in Control

Reviewing

Ad-Hoc Reviewer (journals in alphabetical order)
Acta Psychologica
Cognition
Journal of Experimental Psychology: Human Perception and Performance
Memory & Cognition
Psychological Research
Psychonomic Bulletin & Review

Service Activities

Faculty Search Committee Member, Psychological & Brain Sciences, Washington University in St. Louis, 2024-2025
Co-chair, McDonnell Academy Panel of Scholars (MAPS), 2024-2025
Committee Member, McDonnell Academy Panel of Scholars (MAPS), 2023-2026
Department Diversity Committee, Washington University in St. Louis, 2022-2023
Social Media Committee, Psychology Graduate Student Association, 2022-2023
Peer Mentoring Committee, Psychology Graduate Student Association, 2022-2023